

C > **Proteome extraction**

□  R0151 Urea lysis buffer, 50 mL

- (1.) Add 5 vol ice-cold 1 × urea lysis buffer with inhibitors, about 100 µL per 1×10^7 cells or 5 mg of tissue. Pipette the sample up and down to break up cell clumps and vortex gently.
- (2.) Boil the lysate at 95 °C for 5 min.
- (3.) *Optional:* Sonicate the lysate twice for 10 s at 20 kHz on ice to disrupt cellular membranes, shear genomic DNA, and aid with protein solubilization.

Critical: Since sonication can damage proteins, minimize pulse length and number of pulses. Apply short pulses and lower the power level to avoid frothing.

- (4.) Remove the insoluble fraction by centrifugation at $12\,000 \times g$ for 20 min at 4 °C.
- (5.) Carefully collect the supernatant containing the soluble protein fraction in a new tube on ice.



⌚ 20 min

 Recipe (available online)  Troubleshooting (available online)  Notes (available online)

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